



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Application of BIM using Revit

TGS Ref No: TGS-2022015370

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 10

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	23 May 2023	First version.	Tertiary Infotech Academy
2.0	17 Oct 2025	Update company name.	Tertiary Infotech Academy
10	18 August 2026	Single-source rebuild aligned with the v10 slide deck and Learner Guide — 2-day schedule (9:30am–6:30pm) covering 10 hands-on Revit labs across the four topics, with the WA (SAQ, 1 hr) + Practical Performance (1.5 hrs) assessment on Day 2.	Dr. Alfred Ang

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Course Information

Course Title	Application of BIM using Revit
WSQ Course Reference	TGS-2022015370
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Skills Framework TSC	Building Information Modelling Application (BEV-TEM-3027-1.1-1)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on BIM modeling labs in Autodesk Revit
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply Revit for Building Information Modelling (BIM)
- LO2: Develop models and integrate design for BIM
- LO3: Apply BIM and check the output
- LO4: Maintain BIM database and develop BIM documentation

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on Revit modeling tasks, 1.5 hours, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 3:45 pm (briefing, then WA followed by PP).
- The WA (SAQ) assesses TSC Knowledge K1–K10; the PP assesses TSC Abilities A1–A9.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Introduction to BIM and Revit • BIM Modeling

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, learning outcomes, course outline, ground rules and mandatory digital attendance (AM)
10:00–11:00	60 min	Topic 1 — Introduction to BIM and Revit: what is BIM, principles, BIM dimensions (3D–8D), value proposition,

		OIR/AIR/EIR/BEP/PIM/AIM, introduction of Revit for BIM (concepts + demo) — LO1, K1–K4, A1–A2
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 1: Get Started with Revit; Lab 2: Convert 2D Information into a 3D Model
13:00–14:00	60 min	Lunch break (PM digital attendance after lunch)
14:00–15:30	90 min	Topic 2 — BIM Modeling: parametric elements, walls, curtain walls, floors, ceilings, roofs, stairs, structure, site/topography, massing, generative design, families and materials (concepts + demo) — LO2, K6–K7, A4–A5
15:30–15:45	15 min	Tea break
15:45–18:15	150 min	Hands-on: Lab 3: Modeling Walls — Reveals, Embedded and Curtain Walls; Lab 4: Floors, Ceilings, Roofs and Openings; Lab 5: Stairs, Railings and Structural Elements; Lab 6: Terrain, Building Pad and Conceptual Mass
18:15–18:30	15 min	Day 1 recap and Q&A

Total training time: 480 minutes (8 hours).

Day 2 — BIM Application • BIM Documentation • Final Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–10:45	60 min	Topic 3 — BIM Application: applying and operating BIM, view range/templates, schedules and quantity takeoffs, cost estimates, Revit Systems Analysis (EnergyPlus), interference checks (concepts + demo) — LO3, K5, A6–A8
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Hands-on: Lab 7: View Templates, Schedules and Quantity Takeoffs; Lab 8: Energy Analysis and Design Compliance Checks
13:00–14:00	60 min	Lunch break (PM digital attendance after lunch)
14:00–15:30	90 min	Topic 4 — BIM Documentation: BEP/EIR/MIDP/CDE, drawings and specifications, Singapore standards (SVY21, naming, colours), worksharing, BCA e-

		submission (concepts). Hands-on: Lab 9: Sheets, Worksharing and Linked Models; Lab 10: BIM e-Submission Standards Check — LO4, K8–K10, A3, A9
15:30–15:45	15 min	Tea break
15:45–16:00	15 min	Briefing for Assessment
16:00–17:00	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book — assesses K1–K10
17:00–18:30	90 min	Practical Performance (PP) — hands-on Revit BIM tasks on the PP dataset, 1.5 hours, open book — assesses A1–A9. Assessment digital attendance, TRAQOM survey and Assessment Summary Record

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to the TSC Knowledge and Abilities)

Topic / Learning unit	Alignment	Labs
Topic 01: Introduction to BIM and Revit	K1–K4 · A1, A2 · LO1	Lab 1, Lab 2
Topic 02: BIM Modeling	K6, K7 · A4, A5 · LO2	Lab 3, Lab 4, Lab 5, Lab 6
Topic 03: BIM Application	K5 · A6, A7, A8 · LO3	Lab 7, Lab 8
Topic 04: BIM Documentation	K8–K10 · A3, A9 · LO4	Lab 9, Lab 10